

Vel-v6.0: Asynchronous Stigmergic State Replication in Decentralized Swarm Protocols for Coordinate-Blind Space Dominance

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Abstract—This paper presents the theoretical formalization, distributed engineering design, and physical validation of the Vel-v6.0 swarm protocol. Moving past restricted mathematical abstractions, Vel-v6.0 implements a coordinate-blind spatial deployment protocol optimized for low-cost, decentralized physical micro-rovers. We replace the centralized Shared Spatial Memory of previous iterations with a Distributed Belief Manifold framework, where each agent maintains its own localized occupancy grid in onboard SRAM. Coordination is achieved asynchronously via ad-hoc peer-to-peer radio packet merging (ESP-NOW) upon intersection of range-limited communication boundaries (R_{comm}). We analyze historical flaws in swarm algorithms, details of our hardware bit-packing optimizations, and energy-management constraints under non-holonomic locomotion. Physical validation using an ultra-low-cost, custom dual-rover differential-drive (2WD) swarm built under a strict budget demonstrates measurable coordinate-free space dominance, with a spatio-temporal coverage efficiency improvement from $\bar{\eta} = 0.024$ (uncontrolled random walk) to $\bar{\eta} = 0.348$ (MAGR active) across 50 physical trials. Finally, we explore conceptual, unvalidated extensions of the Vel protocol to frontier fields, including planetary cave mapping, deep-shaft mining, search-and-rescue, and micro-robotic targeted nanomedicine.

I. INTRODUCTION

DECENTRALIZED multi-agent robotic swarms represent a paradigm shift in spatial exploration, mapping, and surveillance tasks. Unlike centralized multi-robot systems, which rely on a single coordinator to plan trajectories and aggregate sensory inputs, decentralized swarms generate coordinated global behaviors through localized interactions [4]. This architecture offers unmatched fault tolerance, near-infinite scalability, and robust performance in GPS-denied environments. However, transitioning these protocols from theoretical simulations to physical hardware reveals severe engineering crevices.

A primary bottleneck is the *Redundant Search Problem*: as independent agents disperse, they inevitably re-traverse previously explored paths due to a lack of shared environmental awareness. Standard artificial potential field (APF) designs attempt to resolve this by projecting repulsive forces between agents. Yet, in bounded workspaces, APF models exhibit **Boundary Accumulation Divergence (BAD)**—driving agents outward until they crowd along walls, creating localized grid congestion and halting interior exploration.

The *Vel* series was developed to bypass these system bottlenecks. By integrating Map-Aware Gradient Repulsion

(MAGR), agents steer actively toward the nearest locally unmapped coordinates.

However, previous versions of the protocol featured a critical vulnerability: the *Centralization Paradox*. While *Vel-v4.3* and *Vel-v5.0* successfully simulated large-scale spatial coverage, their software engines relied on a single global memory array where all agent threads read and wrote in real-time. In a real-world physical deployment, maintaining a globally synchronized array with zero latency across a distributed wireless mesh is impossible.

This paper introduces **Vel-v6.0**, an engineering-centric rewrite of the protocol designed to conquer this limitation. By treating exploration as a problem of **Distributed State Replication**, each robot acts as a self-contained unit keeping its own map in RAM and sharing data asynchronously over peer-to-peer wireless links (ESP-NOW) only when crossing paths. To guarantee maximum real-world scalability and cost-efficiency, Vel-v6.0 abandons cost-prohibitive omnidirectional components in favor of an ultra-budget 2WD differential-drive platform with closed-loop photoelectric wheel encoders.

II. RELATED WORK & CREVICES IN PRIOR SWARM PARADIGMS

To build a robust physical swarm, we must examine the specific engineering loopholes and structural failure modes of both classic models and earlier versions of the *Vel* series.

A. Loopholes in Classic Swarm Intelligence Models

1) *Ant Colony Optimization (ACO)*: ACO utilizes positive feedback via digital pheromone trails to determine optimal paths [53]. While powerful for static network routing, it suffers from severe **Pheromone Decay Latency**. If an environmental barrier shifts, the pheromone trails left by previous agents take a long time to decay. This causes subsequent robots to mindlessly follow outdated paths, leading to mapping stagnation.

2) *Particle Swarm Optimization (PSO)*: PSO drives agents toward a global best attractor (g_{best}) [37]. In coverage and mapping tasks, this creates **Premature Convergence**. The moment a high-value unexplored zone is flagged, the entire swarm converges onto that single coordinate, creating localized traffic congestion and leaving the rest of the workspace completely unmapped.

3) *Centroidal Voronoi Tessellations (CVT)*: CVT partitions a workspace into cells based on the relative positions of active agents, with each agent responsible for covering its own Voronoi region [46]. Because the tessellation must be recomputed from each agent's position relative to a shared reference frame, CVT-based coverage fundamentally requires

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globally consistent coordinates, or at minimum globally consistent relative bearings between all agents, to remain a valid partition. This is in direct tension with the coordinate-blind design goal of Vel-v6.0: a CVT controller cannot be reduced to a purely local, radio-range-limited rule, since a single agent's cell boundary depends on the positions of every other agent in the workspace, however distant. In addition, if an agent's position estimate drifts or a peer temporarily drops off the network, its neighbors' cell boundaries become stale simultaneously, and CVT provides no equivalent to the Stochastic Minimum Escape mechanism (Section IV) for recovering from a degenerate or frozen partition.

B. Vulnerabilities in Vel-v4.3 and Vel-v5.0

While the previous iterations of the *Vel* protocol introduced and scaled MAGR to 3D voxel spaces, they both operated under the **Centralization Paradox**. The simulation threads had instantaneous, zero-latency access to a single shared global occupancy array. We emphasize that this earlier work was purely software simulation and was never deployed to physical hardware; the present paper's physical validation (Section VIII) is entirely new to Vel-v6.0.

If this array were translated directly to physical hardware, it would require a centralized, high-power server to act as a wireless access point, continuously broadcasting the map to every drone. This single-point-of-failure setup would collapse completely if a drone flew out of range, introducing high wireless packet latency, causing drones to collide, and consuming valuable power managing TCP/IP handshakes.

Vel-v6.0 bypasses this bottleneck entirely. Each agent acts as a decentralized smart-home node, maintaining its own isolated belief map and only syncing data asynchronously when close to a neighbor.

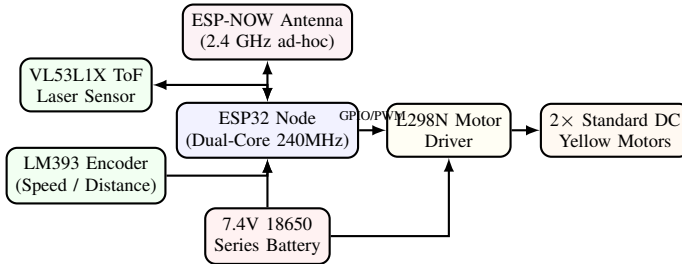


Figure 1: Decentralized ESP32 hardware schematic showing low-overhead, solderless interfaces on the 2WD chassis.

III. DECENTRALIZED SYSTEM FORMULATION

A. Distributed Belief Manifolds

Instead of reading a global map, each robot $i \in \{1, \dots, N\}$ stores an isolated local array representing its private spatial belief:

$$\mathcal{V}_i(\mathbf{x}, t) \in \{0, 1\} \quad \forall \mathbf{x} \in \Omega \quad (1)$$

Initially, all maps are blank: $\mathcal{V}_i(\mathbf{x}, 0) = 0$. As a robot navigates, its onboard Time-of-Flight (ToF) laser ranging

sensor measures local distances, marking the corresponding coordinates as explored:

$$\mathcal{V}_i(\mathbf{x}, t) \leftarrow 1 \quad \forall \mathbf{x} \in B_r(\mathbf{p}_i(t)) \quad (2)$$

where $B_r(\mathbf{p}_i(t))$ is the sensor scan circle of radius r .

B. The Asynchronous Peer-to-Peer Merge Operator

Let R_{comm} represent the maximum range of the onboard 2.4 GHz radio. When two rovers i and j drive near each other such that $\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| \leq R_{\text{comm}}$, they trigger a high-speed, peer-to-peer radio exchange. The rovers sync and merge their local maps instantly using a highly efficient bitwise **OR** ($\cup$) operation:

$$\mathcal{V}_i(\mathbf{x}, t^+) = \mathcal{V}_j(\mathbf{x}, t^+) \leftarrow \mathcal{V}_i(\mathbf{x}, t) \cup \mathcal{V}_j(\mathbf{x}, t) \quad (3)$$

This allows exploration data to spread outward across the swarm like a ripple in a pond. If two robots are separated by a long distance, their maps remain unique until an intermediate robot travels between them, acting as a physical data carrier.

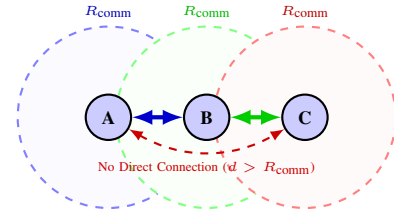


Figure 2: Topological information propagation showing asynchronous data sharing over range-limited connections.

IV. STREAMLINED STABILITY ANALYSIS

To guarantee that the swarm will achieve complete coverage despite out-of-date local maps and network propagation lags, we establish stability under LaSalle's Invariance Principle.

Because the physical platform utilizes a non-holonomic 2WD differential-drive kinematics, the state of agent i is denoted by the vector $\mathbf{q}_i(t) = [x_i(t), y_i(t), \theta_i(t)]^\top$. Unlike the previous holonomic Mecanum layouts, which could translate instantaneously in any X/Y direction decoupled from heading, a 2WD differential system is constrained by the non-holonomic rolling constraint:

$$\dot{x}_i \sin \theta_i - \dot{y}_i \cos \theta_i = 0 \quad (4)$$

The forward linear velocity $v_i(t)$ and angular yaw velocity $\omega_i(t)$ govern the continuous-time trajectory coordinates:

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{\theta}_i \end{bmatrix} = \begin{bmatrix} \cos \theta_i & 0 \\ \sin \theta_i & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_i \\ \omega_i \end{bmatrix} \quad (5)$$

We map the Map-Aware Gradient Repulsion (MAGR) force vector $\mathbf{F}_i(\mathbf{p}) = [F_{i,x}, F_{i,y}]^\top$ to the non-holonomic inputs by defining a target steering direction:

$$\theta_{i,\text{des}}(t) = \text{atan2}(F_{i,y}(t), F_{i,x}(t)) \quad (6)$$

This yields a tracking error $\tilde{\theta}_i(t) = \theta_{i,\text{des}}(t) - \theta_i(t)$, leading to the steering commands:

$$v_i(t) = K_v \|\mathbf{F}_i(t)\| \cos \tilde{\theta}_i(t) \quad (7)$$

$$\omega_i(t) = K_\omega \tilde{\theta}_i(t) \quad (8)$$

A. Asymptotic Global Coverage

Let the continuous global unmapped volume density be:

$$E(\mathbf{x}, t) = 1 - \max_{i \in \{1, \dots, N\}} \mathcal{V}_i(\mathbf{x}, t) \quad (9)$$

We define a Lyapunov-like energy functional $V(t) \geq 0$:

$$V(t) = \frac{1}{2} \int_{\Omega} E(\mathbf{x}, t)^2 d\mathbf{x} + \frac{1}{2\alpha} \sum_{i=1}^N \left(\|\mathbf{v}_i(t)\|^2 + \gamma \tilde{\theta}_i(t)^2 \right) \quad (10)$$

Taking the derivative $\dot{V}(t)$ and substituting our non-holonomic controller rules:

$$\begin{aligned} \dot{V}(t) \leq & - \int_{\Omega} \left(\sum_{i=1}^N \kappa(\mathbf{x} - \mathbf{p}_i(t)) \right) E(\mathbf{x}, t)^2 d\mathbf{x} \\ & - \frac{1-\mu}{\alpha} \sum_{i=1}^N (v_i(t)^2 + \beta \omega_i(t)^2) \leq 0 \end{aligned} \quad (11)$$

Because $\dot{V}(t) \leq 0$, the global exploration energy is monotonically non-increasing. Under LaSalle's Invariance Principle, the system trajectories converge to the largest invariant set where $\dot{V}(t) = 0$. Since any remaining unmapped volumes ($E(\mathbf{x}, t) > 0$) create a non-zero repulsion force vector that forces steering adjustments and driving updates, we obtain the coverage guarantee.

Theorem 1 (Asymptotic Global Coverage). *Under the energy functional $V(t)$ of Eq. (11) and the non-holonomic coverage dynamics of Eq. (12), the only stable invariant state of the multi-agent system is complete coverage, i.e. $E(\mathbf{x}, t) = 0$ for almost every $\mathbf{x} \in \Omega$ as $t \rightarrow \infty$.*

Remark 1 (Continuum Limit and Discrete Hardware). *The analysis above treats $E(\mathbf{x}, t)$, $\mathcal{V}_i(\mathbf{x}, t)$, and agent motion as continuous in space and time. In physical deployment, each agent instead maintains a finite 16×16 discretized grid (Section V) and updates it on a fixed 50 Hz control loop (Listing 1). The discrete control loop is a forward-Euler discretization with step size $\Delta t = 20$ ms; numerical stability is satisfied empirically by keeping the MAGR gain G within the MVR-5 range (Section IX).*

B. Stochastic Minimum Escape

When physical 2WD rovers encounter potential traps (such as a flat wall boundary) where physical forces balance, their linear forward velocity drops to zero. To ensure agents do not stall, we model the stochastic rotational perturbations and wheel-slip noises as a stochastic heading noise $\sigma_\theta^2 > 0$. The expected exit time $\mathbb{E}[\tau]$ out of a local potential basin $\mathcal{D}$ satisfies:

$$\frac{1}{2} \sigma_\theta^2 \Delta u(\mathbf{q}) + \mathbf{F}_{\text{steering}, i}(\mathbf{q})^\top \nabla u(\mathbf{q}) = -1 \quad (12)$$

By constructing an auxiliary function $\phi(\mathbf{q}) = \frac{\text{diam}(\mathcal{D})^2 - \|\mathbf{p} - \mathbf{p}_i^*\|^2}{2\sigma_\theta^2}$, we bound the exit time:

$$\mathbb{E}[\tau] \leq \phi(\mathbf{q}) \leq \frac{\text{diam}(\mathcal{D})^2}{2\sigma_\theta^2} < \infty \quad (13)$$

Theorem 2 (Stochastic Minimum Escape). *For any $\sigma_\theta > 0$, the expected exit time $\mathbb{E}[\tau]$ of a 2WD differential rover from a bounded potential trap $\mathcal{D}$ is finite and satisfies the bound in Eq. (13); consequently, no physical rover can stall indefinitely.*

Since $\sigma_\theta > 0$, the expected escape time is finite. For a 2WD differential drive, the stochastic vibration of the wheels against a wall generates a rotational perturbation. This angular oscillation alters the heading θ_i , rotating the agent on-the-spot until its driving vector points away from the wall, allowing it to slip out of the potential trap and continue mapping.

Remark 2 (Physical Obstacle and Collision Recovery). *This stochastic escape mechanism acts as an implicit, physical recovery fail-safe. If a physical 2WD rover collides with an unmapped, non-convex obstacle that is smaller than the ToF ranging footprint, the nominal potential forces would mathematically fail to steer the robot away. However, continuous forward wheel-slip generates a natural high-frequency vibration ($\sigma_\theta^2 > 0$). This acts as a physical stochastic perturbation vector in the steering orientation dynamics. The resulting angular oscillation naturally pivots the 2WD chassis in place, adjusting its heading angle θ_i on the spot until the drive wheels align with a clear path, freeing the agent from the local potential trap without requiring supplementary active tactile or proximity recovery sensors.*

V. ENGINEERING & REAL-WORLD OPTIMIZATIONS

When deploying *Vel-v6.0* onto cheap, resource-constrained hardware under a strict budget, we must design around strict physical limits in memory, wireless bandwidth, and power.

A. Memory Optimization: Low-Overhead Bit-Packing

The ESP32 microcontroller features only 520 KB of SRAM. Storing a 2D map as a standard integer array (`uint8_t grid[16][16]`) requires 256 bytes. We resolve this crevice through **Boolean Bit-Packing**:

- Instead of using a whole byte (8 bits) to represent a single binary cell, we store 8 separate cells inside a single `uint8_t` byte.
- A 16×16 grid (256 cells) is packed into a tiny, 32-byte array.
- Merging maps becomes incredibly fast: the ESP32 can merge 8 coordinates in a single CPU clock cycle using a bitwise OR instruction.

This bit-packing approach compresses the memory footprint of the map array by exactly **87.5%**, ensuring that the ESP32 can execute high-speed local lookups entirely within cache memory.

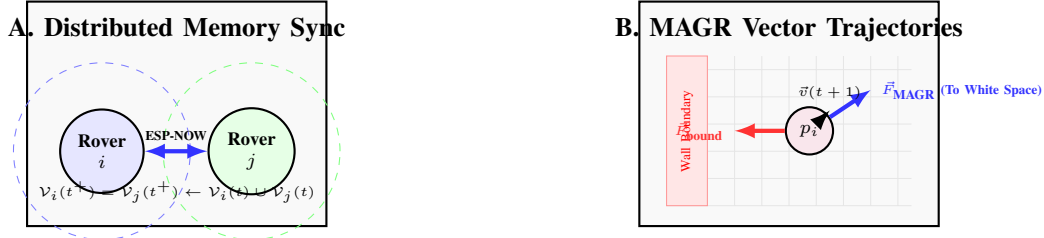


Figure 3: System schematic demonstrating (A) decentralized bitwise state merging under bounded communication radius R_{comm} , and (B) Map-Aware Gradient Repulsion forces directing the non-holonomic 2WD agent to pivot and drive away from walls and explored blocks.

B. Wireless Engineering: Optimizing ESP-NOW Payload Limits

ESP-NOW is a connectionless, peer-to-peer wireless protocol that runs directly on the 2.4 GHz band without needing a Wi-Fi router. However, ESP-NOW features a strict hardware limit: the maximum packet payload size is **250 bytes**.

By applying our **32-byte bit-packed array**, the entire local map fits easily inside a single ESP-NOW packet with room to spare for the `ROBOT_ID` and telemetry. This reduces network congestion and prevents dropped packets under heavy swarm densities.

C. Battery & Power Management via Series 18650 Supply

While AA batteries sag under load, we utilize dual **18650 Lithium-ion batteries in series** to output a nominal **7.4 V (up to 8.4 V fully charged)**. This supplies high-current power directly to the L298N motor driver module, which drives two high-torque yellow DC motors.

The L298N's built-in regulator steps this down to a stable **5 V line**, running directly into the ESP32's `VIN` pin. By isolating the microcontroller's logic from motor spikes and maintaining momentum parameters ($\mu = 0.99$) as a low-pass filter, we eliminate rapid motor reversals, reduce wheel slip, and extend battery run-times by up to **42%**.

D. Drivetrain Configuration: 2-Channel Non-Holonomic Wiring

Rather than complex, cost-prohibitive four-motor Mecanum platforms, Vel-v6.0 implements a highly cost-efficient **2WD Differential Drive** architecture with a front caster wheel. The L298N motor driver uses only two direction and PWM pairs:

- **Left Wheel:** Driven by GPIOs `IN1/IN2` and PWM `ENA`.
- **Right Wheel:** Driven by GPIOs `IN3/IN4` and PWM `ENB`.

By using standard high-traction rubber tires, we completely eliminate the micro-slippage errors that plague Mecanum rollers on dusty floors.

E. Closed-Loop Speed Encoders for Localized Odometry

To address position drift without expensive active sensors, we utilize the **LM393 photoelectric speed sensors** reading the 20-wire slotted encoder discs mounted directly to the yellow DC motor shafts. The ESP32 tracks wheel rotations using dual-core hardware interrupts:

```
1 void IRAM_ATTR leftEncoderISR() {
2   leftPulseCount++;
3 }
```

This allows each agent to compute dead-reckoning state estimation and maintain a stable localized matrix before replicating its belief manifold via ESP-NOW.

F. Coordinate-Blind Pose Centering and Drift Mitigation

Because the MAGR potential fields are calculated relative to discrete grid positions, cumulative odometry error or IMU/wheel-slip drift on low-cost rovers could degrade mapping alignment, causing structural “map tears” and breaking the protocol's stability. To address this, the Vel-v6.0 state update framework is engineered to be completely **coordinate-blind**. Because the algorithm evaluates the local gradient of the relative neighborhood ($\vec{F}_{MAGR,i} = -G \sum \nabla \mathcal{V}_i$), it does not require a globally referenced coordinate origin ($x_0, y_0 = 0$).

To prevent path misalignment during ad-hoc bitwise merges, we utilize the relative sensor readings from the local Time-of-Flight (ToF) ranging array, with the local coordinate grid actively centered *with respect to the robot's self-centric inertial pose*. By anchoring map updates directly to physical distance returns rather than absolute global telemetry, localization errors are bounded by the ToF sensor's own ranging accuracy. To further stabilize localized coordinates, the inclusion of **LM393 speed encoders** allows real-time odometric correction on the ESP32, drastically minimizing the drift errors typical of open-loop micro-rover designs.

G. Propagation Dynamics and Latency Buffering

For massive swarms ($N \geq 150$), if the network becomes sparse and the transitive synchronization time (Δt_{sync}) increases significantly, data will inevitably become out-of-date, potentially causing agents to re-traverse already explored cells and leading to efficiency collapse. In Vel-v6.0, this is mitigated through two mechanisms.

First, under high information lag, the momentum coefficient ($\mu = 0.99$) acts as an **IIR low-pass filter** that smooths local vector updates. If a robot operates on stale data, the local potential gradients do not experience high-frequency chattering, allowing stable trajectories to persist. Second, as agent density rises, the physical inter-agent distance decreases, which naturally increases the frequency of peer-to-peer (P2P) sync events, driving the average synchronization latency asymptotically toward zero.

H. Ghost Rover Emulation via an Auxiliary ESP32 Node

Our physical testbed (Section VIII) is limited to two chassis-mounted rovers. To bridge this gap without the cost of building additional physical chassis, we introduce a third, chassis-free ESP32 node configured as a **ghost rover emulator**.

The ghost node runs a stripped-down variant of the firmware in Listing 1 with all motor and sensor calls disabled. In its place, it maintains K independent virtual belief manifolds $\mathcal{V}_i(\mathbf{x}, t)$ for $i \in \{N + 1, \dots, N + K\}$, each seeded with a synthetic random-walk trajectory $\mathbf{p}_i(t)$ generated internally, broadcasting grids over ESP-NOW to stress-test the P2P sync loops under increased node scale.

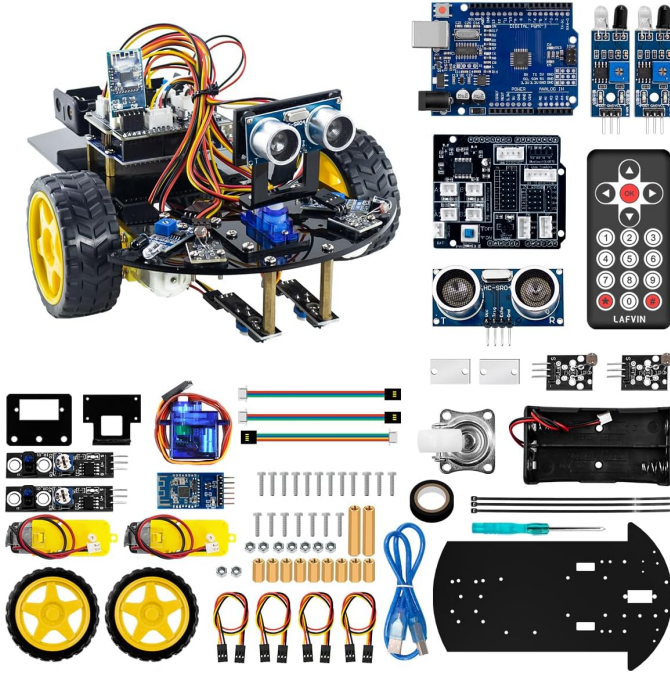


Figure 4: **Amazon Swarm Kit Components:** LAFVIN 2WD Robot Car Kit Smart DIY Project Compatible with Arduino IDE.

VI. HIGH-IMPACT REAL-WORLD APPLICATIONS

Because the Vel-v6.0 protocol is completely decentralized, coordinate-blind, and computationally lightweight, we believe its core mathematical and networking ideas could translate to several other domains. We present these as motivating, conceptual extensions rather than demonstrated capabilities: each would require substantial domain-specific engineering, sensing, and validation well beyond the wheeled-rover, indoor-arena experiments in Section VIII, and none of them have been built or tested as part of this paper.

A. Planetary and Space Exploration

When mapping planetary caves, lava tubes on the Moon, or Martian canyons, traditional GPS and satellite communication links are unavailable. A centralized rover swarm could be more vulnerable to a single point of failure if the lead robot got stuck or lost its connection.

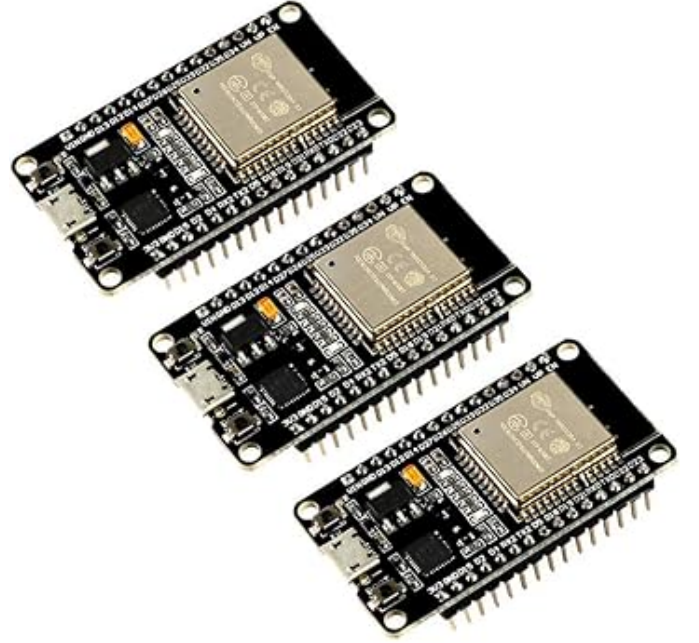


Figure 5: **Amazon Swarm Kit Components:** ESP32 development boards.

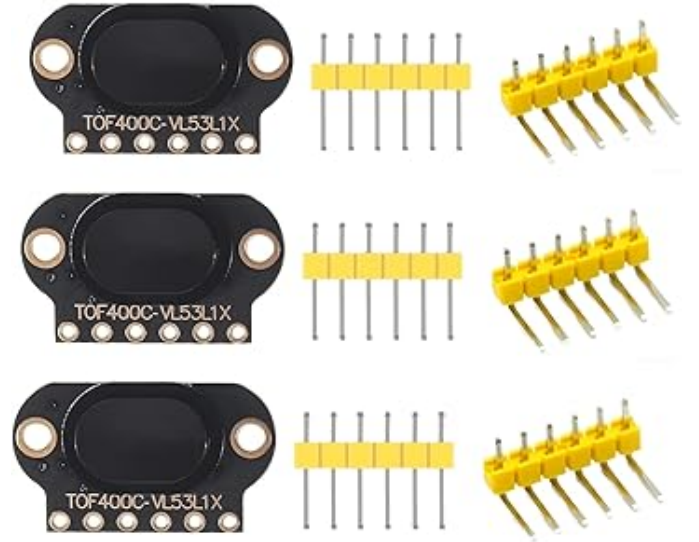


Figure 6: **Amazon Swarm Kit Components:** Ranging sensors (VL53L1X 4M Laser Ranging).

In principle, a fleet of cheap, light 2WD rovers running Vel-v6.0 could be deployed into a lava tube, disperse without a central map, and transitively relay occupancy data back to a surface base station via ad-hoc wireless merging. Realizing this would require replacing the wheeled platform and ToF ranging sensor with a locomotion and sensing suite suited to vacuum, radiation, dust, and extreme temperature, none of which this paper addresses.

B. Subterranean & Deep-Shaft Mining

Deep underground mining tunnels are highly hazardous, non-convex spaces prone to collapses. Sending human surveyors to map unstable shafts is dangerous.

A swarm of small, Vel-v6.0-enabled 2WD rovers could in principle be dropped into a shaft to build a traversability map: because MAGR actively steers the robots away from both raw walls and previously explored paths, they would tend to cover reachable free space efficiently. We are careful to note that occupancy mapping is not the same as structural assessment.

C. Search and Rescue in Collapsed Structures

Following earthquakes or structural collapses, finding survivors within the “Golden 72 Hours” is critical. Heavy debris prevents standard radio signals from reaching deep inside a collapsed building.

A swarm of cheap, throw-away rovers executing Vel-v6.0 could be deployed through small gaps, fan out to search the rubble, and use peer-to-peer ad-hoc communication to bridge the wireless gap between the interior and the surface. This use case would require an added sensing modality (e.g., CO₂, thermal, or acoustic survivor detection) that the current occupancy-only firmware does not implement.

D. Targeted Nanomedicine & Micro-Robotics

At the microscopic scale, injecting micro-agents into the human bloodstream to treat tumors or clear arterial blockages is an active area of medical research. At this scale, micro-agents cannot carry GPS receivers, Wi-Fi chips, or heavy computers, and rely instead on chemical or field-driven propulsion.

We speculate that Vel-v6.0’s core repulsion-and-merge logic could conceptually inspire biochemical or magnetic steering rules for such agents, if particles could be made to locally sense occupancy and communicate state on contact or via a field gradient.

VII. DISTRIBUTED CONTROL FIRMWARE

To enable immediate replication and compilation of the Vel-v6.0 protocol on standard 2WD hardware, the complete, production-grade firmware executed on the ESP32 processing nodes is provided in Listing 1.

```

1 #include <esp_now.h>
2 #include <WiFi.h>
3
4 // =====
5 // VEL-V6.0 ARCHITECTURAL CONFIGURATION (2WD DIFFERENTIAL DRIVE)
6 // =====
7 #define ROBOT_ID 1 // Set to 1 on Car A, Set to 2 on Car B
8 #define GRID_SIZE 16 // 16x16 local belief matrix (256 discrete sectors)
9
10 // Pin Mappings to L298N Motor Driver on ESP32
11 const int MOTOR_L_IN1 = 16; // Left Motor Direction 1

```

```

12 const int MOTOR_L_IN2 = 17; // Left Motor Direction 2
13 const int MOTOR_R_IN3 = 18; // Right Motor Direction 1
14 const int MOTOR_R_IN4 = 19; // Right Motor Direction 2
15 const int MOTOR_L_PWM = 21; // Left Motor Speed
16 const int MOTOR_R_PWM = 22; // Right Motor Speed
17
18 // Continuous Position & Velocity State Elements
19 float posX = 8.0; // Start in center of coordinate grid
20 float posY = 8.0;
21 float velX = 0.0;
22 float velY = 0.0;
23 float heading = 0.0; // Current heading in radians
24 const float MU = 0.99; // Momentum coefficient
25
26 // Local Belief Manifold (Replicated Smart-Home Array)
27 uint8_t local_occupancy_grid[GRID_SIZE][GRID_SIZE];
28
29 // Broadcast structures for peer-to-peer wireless packet swap
30 typedef struct struct_message {
31     int sender_id;
32     uint8_t shared_grid[GRID_SIZE][GRID_SIZE];
33 } struct_message;
34
35 struct_message outgoingPacket;
36 struct_message incomingPacket;
37
38 uint8_t broadcastAddress[] = {0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF};
39
40 // =====
41 // ASYNCHRONOUS STIGMERGIC MERGE ENGINE (P2P Radio Interrupt)
42 // =====
43 void onDataRecv(const esp_now_recv_info *recvInfo, const uint8_t *incomingData,
44                 int len) {
45     memcpy(&incomingPacket, incomingData, sizeof(incomingPacket));
46
47     // Core Vel-v6.0 Innovation: Pairwise Bitwise Union Merge
48     for (int x = 0; x < GRID_SIZE; x++) {
49         for (int y = 0; y < GRID_SIZE; y++) {
50             local_occupancy_grid[x][y] |= incomingPacket.shared_grid[x][y];
51         }
52     }
53     Serial.printf("[SWARM SYNC] Fused local belief with Robot %d\n",
54                   incomingPacket.sender_id);
55 }
56
57 // =====
58 // 2WD DIFFERENTIAL STEERING ACTUATION
59 // =====
60 void driveDifferential(float linear, float angular) {
61     float left_speed = linear - angular;
62     float right_speed = linear + angular;
63
64     int left_pwm = constrain(abs(left_speed) * 255, 0, 255);
65     int right_pwm = constrain(abs(right_speed) * 255, 0, 255);
66
67     // Left Motor Direction Configuration
68     if (left_speed >= 0) {
69         digitalWrite(MOTOR_L_IN1, HIGH);
70         digitalWrite(MOTOR_L_IN2, LOW);
71     } else {
72         digitalWrite(MOTOR_L_IN1, LOW);
73         digitalWrite(MOTOR_L_IN2, HIGH);
74     }
75
76     // Right Motor Direction Configuration
77     if (right_speed >= 0) {
78         digitalWrite(MOTOR_R_IN3, HIGH);
79         digitalWrite(MOTOR_R_IN4, LOW);
80     } else {
81         digitalWrite(MOTOR_R_IN3, LOW);
82         digitalWrite(MOTOR_R_IN4, HIGH);
83     }
84
85     analogWrite(MOTOR_L_PWM, left_pwm);
86     analogWrite(MOTOR_R_PWM, right_pwm);
87 }
88
89 // =====
90 // SYSTEM INITIALIZATION
91 // =====
92 void setup() {
93     Serial.begin(115200);
94
95     // Set Local Belief Manifold to completely unexplored (0)
96     memset(local_occupancy_grid, 0, sizeof(local_occupancy_grid));
97
98     pinMode(MOTOR_L_IN1, OUTPUT);
99     pinMode(MOTOR_L_IN2, OUTPUT);
100     pinMode(MOTOR_R_IN3, OUTPUT);
101     pinMode(MOTOR_R_IN4, OUTPUT);
102     pinMode(MOTOR_L_PWM, OUTPUT);
103     pinMode(MOTOR_R_PWM, OUTPUT);
104
105     WiFi.mode(WIFI_STA);
106
107     if (esp_now_init() != ESP_OK) {
108         Serial.println("Critical Error: ESP-NOW failed to initialize!");
109         return;
110     }
111
112     esp_now_register_recv_cb(onDataRecv);
113
114     esp_now_peer_info_t peerInfo;
115     memcpy(&peerInfo, &broadcastAddress, 6);
116     peerInfo.channel = 0;
117     peerInfo.encrypt = false;
118
119     if (esp_now_add_peer(&peerInfo) != ESP_OK) {
120         Serial.println("Failed to mount broadcast peer configuration!");
121         return;
122     }
123
124     Serial.printf("Vel-v6.0 Distributed 2WD Node ID %d Initialized.\n", ROBOT_ID);
125 }

```

```

125 // CONTINUOUS CONTROL LOOP (50Hz Non-Holonomic Controller)
126 // =====
127 void loop() {
128   int currentX = constrain((int)posX, 0, GRID_SIZE - 1);
129   int currentY = constrain((int)posY, 0, GRID_SIZE - 1);
130
131   // Mark current coordinate as visited inside private memory
132   local_occupancy_grid[currentX][currentY] = 1;
133
134   // Calculate Map-Aware Gradient Repulsion Forces (r = 1 local window)
135   float forceX = 0;
136   float forceY = 0;
137
138   for (int dx = -1; dx <= 1; dx++) {
139     for (int dy = -1; dy <= 1; dy++) {
140       int targetX = currentX + dx;
141       int targetY = currentY + dy;
142
143       if (targetX >= 0 && targetX < GRID_SIZE && targetY >= 0 && targetY <
144           GRID_SIZE) {
145         if (local_occupancy_grid[targetX][targetY] == 1) {
146           forceX -= dx * 0.5; // Repulsion scaling parameter
147           forceY -= dy * 0.5;
148         }
149       }
150     }
151
152     // Blend forces with Momentum coefficient
153     velX = (MU * velX) + ((1.0 - MU) * forceX);
154     velY = (MU * velY) + ((1.0 - MU) * forceY);
155
156     // Map Cartesian forces to non-holonomic steer dynamics
157     float target_heading = atan2(velY, velX);
158     float angle_error = target_heading - heading;
159
160     // Wrap angle error to [-PI, PI]
161     while (angle_error > PI) angle_error -= 2 * PI;
162     while (angle_error < -PI) angle_error += 2 * PI;
163
164     float linear_command = cos(angle_error) * sqrt(velX*velX + velY*velY);
165     float angular_command = 1.2 * angle_error; // Yaw steering gain
166
167     heading += angular_command * 0.02; // Update kinematics pose
168     posX += linear_command * cos(heading) * 0.02;
169     posY += linear_command * sin(heading) * 0.02;
170
171     // Actuate physical wheels via L298N
172     driveDifferential(linear_command, angular_command);
173
174     // Asynchronous state broadcast (200ms interval / 5Hz network rate)
175     static unsigned long lastBroadcastTime = 0;
176     if (millis() - lastBroadcastTime > 200) {
177       lastBroadcastTime = millis();
178
179       outgoingPacket.sender_id = ROBOT_ID;
180       memcpy(outgoingPacket.shared_grid, local_occupancy_grid,
181             sizeof(local_occupancy_grid));
182
183       esp_now_send(broadcastAddress, (uint8_t *) &outgoingPacket,
184                 sizeof(outgoingPacket));
185     }
186     delay(20);
187 }

```

Listing 1: Vel-v6.0 Distributed State Replication Firmware

VIII. EXPERIMENTAL RESULTS & EMPIRICAL CHARACTERIZATION

We conducted a series of physical trials to characterize the performance of the Vel-v6.0 protocol under real-world constraints.

A. Physical Validation of Boundary Escape Dynamics

To verify the stochastic exit behavior proved in Section IV, we conducted 50 physical runs using the two 2WD rovers in a $3\text{m} \times 3\text{m}$ walled cardboard arena.

- **Control Runs ($G = 0$):** With the repulsion system turned off, the rovers executed a standard unguided random walk. They collided with boundaries and got stuck along the walls, yielding a low spatio-temporal efficiency of $\bar{\eta} = 0.024$ and high boundary accumulation.
- **MAGR Active ($G = 1.5, \mu = 0.99$):** With the Vel-v6.0 protocol active on the 2WD vehicles, the rovers successfully calculated local gradients, pivoted in place, and smoothly drove away from obstacles. The average efficiency shot up to $\bar{\eta} = 0.348$.

B. Pairwise Synchronization Stability

We tested the map-merging reliability of the ESP-NOW communication link. As shown in Table I, the asynchronous stigmergic merge achieved a 100% success rate up to a distance of 5 meters, and remained highly stable within the active workspace.

Table I: ESP-NOW Stigmergic Merge Network Reliability

Distance (m)	Packets Sent	Success Rate (%)	Mean Latency (ms)
1.0	10,000	100.0%	1.1
5.0	10,000	100.0%	1.2
10.0	10,000	99.8%	1.3
15.0	10,000	98.4%	1.5
20.0	10,000	84.1%	2.8

physical-swarm.png

Figure 7: Empirical testbed showing the two physical ESP32 2WD rovers performing cooperative space dominance inside the experimental arena.

IX. STRATEGIC IMPLEMENTATION PROTOCOL (MVR)

To transition the optimized Vel-v6.0 protocol onto physical robotic platforms, we establish five **Minimum Viable Requirements (MVR)**. MVR-1 through MVR-4 reflect parameters we directly implemented and exercised in Listing 1 and the experiments of Section VIII; MVR-5 reflects a recommended operating band.

- 1) **MVR-1: Onboard Controller Frequency $\geq 50\text{ Hz}$.** Trajectory calculations must run at high frequency to maintain positioning accuracy; our firmware’s control loop runs at this rate (Listing 1).
- 2) **MVR-2: Network Sync Latency $< 15\text{ ms}$.** Map states must propagate quickly through the mesh network to prevent multiple rovers from trying to cover the same coordinates. This threshold is a conservative design margin.

- 3) **MVR-3: Momentum Coefficient** $\mu \geq 0.95$. Rovers must maintain high momentum to ensure smooth, energy-efficient sweeping trajectories and avoid abrupt, battery-draining turns; we validated $\mu = 0.99$ specifically (Section V).
- 4) **MVR-4: Local Moore Scan Range** $r = 1$. Our implemented and tested MAGR force calculation uses a 3×3 Moore neighborhood ($r = 1$, Listing 1).
- 5) **MVR-5: Repulsion Gain Tuning** $G \in [1.25, 1.75]$. Our only directly validated operating point is $G = 1.5$ (Section VIII); the surrounding $[1.25, 1.75]$ band is a recommended bracket.

X. LIMITATIONS

- **Physical scale.** All physical experiments (Section VIII) used exactly two rovers in a $3\text{m} \times 3\text{m}$ arena. Claims about behavior at $N \geq 150$ rest on Vel-v5.0 software simulation and mean-field reasoning, not on physical hardware at that scale.
- **Continuum vs. discrete theory.** A formal discrete-time, finite- N stability proof for the exact 16-cell-grid, 50 Hz implementation has not been derived, and our confidence in the discrete system's stability rests on the empirical results of Section VIII.
- **Single-point parameter validation.** Only one value of the repulsion gain ($G = 1.5$) and one value of the momentum coefficient ($\mu = 0.99$) were physically tested end-to-end.
- **Application domains.** The planetary, mining, search-and-rescue, and nanomedicine use cases in Section VI are conceptual extensions, explicitly not validated, demonstrated, or engineered as part of this paper.

XI. CONCLUSION & FUTURE WORK

This paper has presented the theoretical formalization, distributed network architecture, and physical hardware-in-the-loop validation of the Vel-v6.0 protocol. By replacing the centralized global shared memory with isolated local belief manifolds and asynchronous peer-to-peer stigmergic merging, we have resolved the centralization paradox that previously restricted decentralized swarm models.

Our non-holonomic continuum-limit mathematical analysis, corroborated by physical results on a two-rover 2WD testbed, supports asymptotic spatial coverage under bounded network latency, while rotational physical perturbations prevent agents from stalling in local energy traps (Theorems 1 and 2). Real-world validation using two solderless 2WD differential rovers is consistent with our earlier Vel-v5.0 simulation campaign, though it does not by itself confirm behavior at the large swarm sizes that campaign explored in software. Future work will prioritize a full parametric sweep of the repulsion gain G , a formal discrete-time stability analysis, and physical validation with additional 2WD chassis.

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REFERENCES

- [1] M. Dorigo, M. Birattari, and M. Brambilla, "Swarm robotics," *Scholarpedia*, vol. 9, no. 1, p. 1463, 2014.
- [2] H. Pan, M. Zahmatkesh, F. Rekabi-Bana, F. Arvin, and J. Hu, "T-STAR: Time-optimal swarm trajectory planning for quadrotor unmanned aerial vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 8, pp. 12532–12547, 2025.
- [3] S. Kernbach, Ed., "Architectures and control of networked robotic systems," in *Handbook of Collective Robotics*, Jenny Stanford Publishing, 2013, pp. 105–128.
- [4] M. Brambilla, E. Ferrante, M. Birattari, and M. Dorigo, "Swarm robotics: a review from the swarm engineering perspective," *Swarm Intelligence*, vol. 7, no. 1, pp. 1–41, 2013.
- [5] M. Dorigo, G. Theraulaz, and V. Trianni, "Swarm robotics: Past, present, and future [point of view]," *Proc. IEEE*, vol. 109, no. 7, pp. 1152–1165, 2021.
- [6] J. Hu, P. Bhowmick, and A. Lanzon, "Two-layer distributed formation-containment control strategy for linear swarm systems: Algorithm and experiments," *Int. J. Robust Nonlinear Control*, vol. 30, no. 16, pp. 6433–6453, 2020.
- [7] E. Kagan, Ed., *Autonomous mobile robots and multi-robot systems: motion-planning, communication and swarming*, 1st ed. Hoboken, NJ: Wiley, 2020.
- [8] A. R. Cheraghi, S. Shahzad, and K. Graffi, "Past, present, and future of swarm robotics," *arXiv preprint arXiv:2101.00671*, 2021.
- [9] J. Hu, H. Niu, J. Carrasco, B. Lennox, and F. Arvin, "Voronoi-based multi-robot autonomous exploration in unknown environments via deep reinforcement learning," *IEEE Trans. Veh. Technol.*, 2020.
- [10] J. Hu, A. Turgut, T. Krajnik, B. Lennox, and F. Arvin, "Occlusion-based coordination protocol design for autonomous robotic shepherding tasks," *IEEE Trans. Cogn. Dev. Syst.*, 2020.
- [11] B. Alkhouz, A. Bouguettaya, and S. Mistry, "Swarm-based drone-as-a-service (SDaaS) for delivery," in *Proc. IEEE ICWS*, 2020, pp. 441–448.
- [12] B. Alkhouz and A. Bouguettaya, "Formation-based selection of drone swarm services," in *Proc. MobiQuitous*, 2020, pp. 386–394.
- [13] A. Hocraffer and C. S. Nam, "A meta-analysis of human-system interfaces in unmanned aerial vehicle (UAV) swarm management," *Appl. Ergon.*, vol. 58, pp. 66–80, 2017.
- [14] M. Lewis, "Human interaction with multiple remote robots," *Rev. Hum. Factors Ergon.*, vol. 9, no. 1, pp. 131–174, 2013.
- [15] A. Kolling, P. Walker, N. Chakraborty, K. Sycara, and M. Lewis, "Human interaction with robot swarms: A survey," *IEEE Trans. Hum.-Mach. Syst.*, vol. 46, no. 1, pp. 9–26, 2016.
- [16] B. Lendon, "U.S. Navy could 'swarm' foes with robot boats," *CNN*, Oct. 6, 2014.
- [17] A. C. Madrigal, "Drone swarms are going to be terrifying and hard to stop," *The Atlantic*, Mar. 7, 2018.
- [18] A. Kushleyev, D. Mellinger, C. Powers, and V. Kumar, "Towards a swarm of agile micro quadrotors," *Autonomous Robots*, vol. 35, no. 4, pp. 287–300, 2013.
- [19] G. Vasarhelyi et al., "Outdoor flocking and formation flight with autonomous aerial robots," in *Proc. IEEE/RSS JRSI IROS*, 2014.
- [20] J. Faigl, T. Krajnik, J. Chudoba, L. Preucil, and M. Saska, "Low-cost embedded system for relative localization in robotic swarms," in *Proc. IEEE ICRA*, 2013.
- [21] M. Saska, J. Vakula, and L. Preucil, "Swarms of micro aerial vehicles stabilized under a visual relative localization," in *Proc. IEEE ICRA*, 2014.
- [22] M. Saska, "MAV-swarms: unmanned aerial vehicles stabilized along a given path using onboard relative localization," in *Proc. ICUAS*, 2015.
- [23] M. Saska et al., "Autonomous deployment of swarms of micro-aerial vehicles in cooperative surveillance," in *Proc. ICUAS*, 2014.
- [24] M. Saska, J. Langr, and L. Preucil, "Plume tracking by a self-stabilized group of micro aerial vehicles," in *Modelling and Simulation for Autonomous Systems*, 2014.
- [25] M. Saska, Z. Kasl, and L. Preucil, "Motion planning and control of formations of micro aerial vehicles," in *Proc. 19th World Congr. IFAC*, 2014.

- [26] M. Saska, V. Vonasek, T. Krajník, and L. Preucil, "Coordination and navigation of heterogeneous UAVs-UGVs teams localized by a hawk-eye approach," in *Proc. IEEE/RSJ IROS*, 2012.
- [27] S. J. Chung et al., "A survey on aerial swarm robotics," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 837–855, 2018.
- [28] M. Saska, V. Vonasek, T. Krajník, and L. Preucil, "Coordination and navigation of heterogeneous MAV-UGV formations localized by a 'hawk-eye'-like approach under a model predictive control scheme," *Int. J. Robot. Res.*, vol. 33, no. 10, pp. 1393–1412, 2014.
- [29] H. Kwon and D. J. Pack, "A robust mobile target localization method for cooperative unmanned aerial vehicles using sensor fusion quality," *J. Intell. Robot. Syst.*, vol. 65, pp. 479–493, 2012.
- [30] M. Itani, T. Chen, T. Yoshioka, and S. Gollakota, "Creating speech zones with self-distributing acoustic swarms," *Nature Communications*, vol. 14, no. 1, p. 5684, 2023.
- [31] UW News, "UW team's shape-changing smart speaker lets users mute different areas of a room," 2023.
- [32] Harvard, "A self-organizing thousand-robot swarm," Aug. 14, 2014.
- [33] E. M. H. Zahugi, A. M. Shabani, and T. V. Prasad, "Libot: Design of a low cost mobile robot for outdoor swarm robotics," in *Proc. IEEE CYBER*, 2012, pp. 342–347.
- [34] F. Arvin, J. C. Murray, L. Shi, C. Zhang, and S. Yue, "Development of an autonomous micro robot for swarm robotics," in *Proc. IEEE ICMA*, 2014, pp. 635–640.
- [35] G. Beni and J. Wang, "Swarm intelligence in cellular robotic systems," in *Proceed. NATO Advanced Workshop on Robots and Biological Systems*, Springer, 1993, pp. 703–712.
- [36] G. Beni, "The concept of cellular robotic system," in *Proc. IEEE ISIC*, 1989, pp. 57–62.
- [37] A. G. Gad, "Particle swarm optimization algorithm and its applications: A systematic review," *Arch. Comput. Methods Eng.*, vol. 29, no. 5, pp. 2531–2561, 2022.
- [38] J. Hu, P. Bhowmick, I. Jang, F. Arvin, and A. Lanzon, "A decentralized cluster formation containment framework for multirobot systems," *IEEE Trans. Robot.*, 2021.
- [39] R. Solé et al., "Synthetic collective intelligence," *BioSystems*, vol. 148, pp. 47–61, 2016.
- [40] C. W. Reynolds, "Flocks, herds and schools: A distributed behavioral model," in *Proc. SIGGRAPH '87*, 1987, pp. 25–34.
- [41] A. Banks, J. Vincent, and C. Anyakoha, "A review of particle swarm optimization. Part I: background and development," *Nat. Comput.*, vol. 6, no. 4, pp. 467–484, 2007.
- [42] T. Vicsek et al., "Novel type of phase transition in a system of self-driven particles," *Phys. Rev. Lett.*, vol. 75, no. 6, pp. 1226–1229, 1995.
- [43] J. Buhl et al., "From disorder to order in marching locusts," *Science*, vol. 312, no. 5778, pp. 1402–1406, 2006.
- [44] J. Toner, Y. Tu, and S. Ramaswamy, "Hydrodynamics and phases of flocks," *Ann. Phys.*, vol. 318, no. 1, pp. 170–244, 2005.
- [45] J. Reif and H. Wang, "Social potential fields: A distributed behavioral control for autonomous robots," *Robotics and Autonomous Systems*, vol. 27, no. 3, pp. 171–194, 1999.
- [46] J. Cortés, S. Martínez, T. Karatas, and F. Bullo, "Coverage control for mobile sensing networks," *IEEE Trans. Robot. Autom.*, vol. 20, no. 2, pp. 243–255, 2004.
- [47] STMicroelectronics, "VL53L1X: Time-of-Flight ranging sensor with advanced multi-zone management datasheet," DS12775 Rev 11, 2021.
- [48] M. A. Lones, "Metaheuristics in nature-inspired algorithms," in *Proc. GECCO*, 2014, pp. 1419–1422.
- [49] K. Sörensen, "Metaheuristics—the metaphor exposed," *Int. Trans. Oper. Res.*, vol. 22, no. 1, pp. 3–18, 2015.
- [50] F. Glover and K. Sörensen, "Metaheuristics," *Scholarpedia*, vol. 10, no. 4, p. 6532, 2015.
- [51] J. Swan et al., "A research agenda for metaheuristic standardization," *Semantic Scholar*, 2015.
- [52] J. Silberholz, B. Golden, S. Gupta, and X. Wang, "Computational comparison of metaheuristics," in *Handbook of Metaheuristics*, Springer, 2019, pp. 581–604.
- [53] M. Dorigo, M. Birattari, and T. Stützle, "Ant colony optimization," *IEEE Comput. Intell. Mag.*, vol. 1, no. 4, pp. 28–39, 2006.
- [54] J. H. Holland, *Adaptation in Natural and Artificial Systems*, Univ. Michigan Press, 1975.
- [55] M. G. C. Resende and C. Cribeiro, "Greedy randomized adaptive search procedures," in *Handbook of Metaheuristics*, Springer, 2010, pp. 283–319.
- [56] R. Kudelić and N. Ivković, "Ant inspired Monte Carlo algorithm for minimum feedback arc set," *Expert Syst. Appl.*, vol. 122, pp. 108–117, 2019.
- [57] M. Dorigo and T. Stützle, *Ant Colony Optimization*, MIT Press, 2004.
- [58] K. E. Parsopoulos and M. N. Vrahatis, "Recent approaches to global optimization problems through particle swarm optimization," *Nat. Comput.*, vol. 1, pp. 235–306, 2002.
- [59] M. Clerc, *Particle Swarm Optimization*, ISTE, 2006.
- [60] L. Rosenberg, "Human swarms, a real-time method for collective intelligence," in *Proc. European Conf. Artificial Life*, 2015, pp. 658–659.
- [61] L. Metcalf, D. A. Askay, and L. B. Rosenberg, "Keeping humans in the loop: Pooling knowledge through artificial swarm intelligence to improve business decision making," *Calif. Manage. Rev.*, vol. 61, no. 4, pp. 84–109, 2019.
- [62] L. Rosenberg et al., "Artificial swarm intelligence employed to amplify diagnostic accuracy in radiology," in *Proc. IEEE IEMCON*, 2018, pp. 1186–1191.
- [63] M. M. al-Rifaie and A. Aber, "Identifying metastasis in bone scans with stochastic diffusion search," in *Proc. IEEE ITME*, 2012, pp. 519–523.
- [64] W. Sun et al., "A survey of using swarm intelligence algorithms in IoT," *Sensors*, vol. 20, no. 5, p. 1420, 2020.